Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element G Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your construction of a testable prototype, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own prototype construction details, be sure to delete the instructions.

Guidance before filling out the template that follows:

- ➔ Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of this Element as another piece of a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element also.
- ➔ You will then proceed on to test and analyze this design and then determine your next iteration, returning to this Element afterwards when you are ready to construct a new testable prototype. Fill out each of these steps again.

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation *(G1)*

This section focuses on the process of explaining your prototypes. While the title of the subelement focuses on your final prototype, you must have at least an initial prototype and a final prototype, with additional prototypes in-between being desirable. Note that you never know which prototype is your final prototype until after you test it, so be sure to record all of the information for each prototype. Use as many copies of this sub-template as needed.

Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

Name: ______________________________ **Date:** _________________ **Class:** _______________

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date: 3/14

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

3/14	Details	Tools and Manufacturing Methods needed
Step 1	Sort the materials	Drill with drill bits
Step 2	Sort the materials	Impact driver with bit for screws
Step 3	Sort the materials	Screws

Name: ______________________________ **Date:** _________________ **Class:** _______________

Step 4	Sort the materials	Slip joint pliers.
Step 5	Sort the materials	Sort the materials
Step 6	Sort the materials	Sort the materials
Step 7	Sort the materials	Sort the materials

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #2 (change this number to match the design number you're on)

Date:
Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Name: ______________________________ **Date:** _________________ **Class:** _______________

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

3/18	Details	Tools and Manufacturing Methods used
Step 1	Measure and cut the wood	Hand saw, tape measure, and sharpie
Step 2	Drill the cut up wood together	Drill gun and screws
Step 3	Measure and plan where the holes on the pvc pipe will go	Tape measure and sharpie
Step 4		

Name: ______________________________ **Date:** _________________ **Class:** _______________

Step 5		
Step 6		
Step 7		

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	**Description**	**Where material was purchased**	**Cost per unit**	**# units**	**Total cost per item**
Pvc pipe	To have the water transported through the pipe	Ralph	N/A	1	N/A
1x2x8ft wood	The wood is the stand/support to elevate the Pvc pipe to over the land needed to be watered	Home depot	$1.53	2	$3.06

Name: ______________________________ **Date:** _________________ **Class:** _______________

Digital timer outlet	Water timer for the irrigation system to be automatic	Amazon	$15.99	1	$15.99
Electric Utility Sump Transfer Water Pump	The pump is to make the water pressure higher for the water to meet the end of the irrigation system	Amazon	$49.99	1	$49.99
Slip cap fitting	To cap the end of the pvc, so the water does not flow out of the end.	Amazon	$1.39	1	$1.39
Sealant tape	To wrap the threads of the hoses and pipes to other parts without the water leaking out.	Amazon	$3.50	1	$3.50
PVC Hose Adapter with 3/4-Inch Female Hose	This slips on the PVC and allows us to thread the	Amazon	$12.07	1	$12.07

Name: ______________________________ **Date:** _________________ **Class:** _______________

	garden hose onto the PVC.				
Hose to Hose Swivel, 3/4-Inch	This will connect the provided hose coming off the spigot and the pump.	Amazon	$3.05	1	$3.05

Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype *(G2)*

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Must be automatic	The prototype will be connected to a programmable automatic timer
Must last 10 days	The prototype will be self sustainable

Name: ______________________________ **Date:** _________________ **Class:** _______________

Must be connected to a water barrel	The prototype will be connected to the water barrel because the water barrel will be the source of water for the irrigation system
Must be portable	This prototype allows the group to collect the objective and measurable data by being portable to observe if the irrigation system can face transportation and being in a new environment

Testing or Modeling of Attributes *(G3, G4)* and Addressing Non-Testable Attributes *(G4)*

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter "expert review required" for these attributes in the final column.

Attribute	**Description**	**Describe your prototype's functionality to test this attribute**
Raised off the ground	The irrigation system is raised off the ground for the entire land to get the	The prototypes functionality to test this attribute is to show that the entire area

Name: ______________________________ **Date:** _________________ **Class:** _______________

	proper amount of water a day	will be receiving water
Has a water pump	Between the barrel and hose there will be a powered pump that will allow for water pressure to spread the water among the garden	The prototype will include a water pump for the purpose of adding water pressure
Water timer	The water timer is to make the prototype automatic	The water timer will make the irrigation system able to water to land on its own without the assistance of a secondary element
Many holes in the PVC pipe	The pvc pipe is the main component for the irrigation system to be able to spread the water through the entire area through the holes	The purpose of the various holes in the PVC is so that we can evenly spread the water throughout the garden

Addressing Non-Testable Attributes *(G4)*

For all attributes that you listed “expert review required” in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	**Justification for why this attribute cannot be modeled or tested with your prototype**
Water barrel	The water barrel is not able to be tested because the water barrel is given and will not affect the result of the irrigation system

Name: ______________________________ **Date:** _________________ **Class:** _______________

The environment that the system is in	Our prototype will not actually be placed over Dr. Jacksons garden. We will have to test it at CHS.